

LABORATORY OBSERVATION

Course: Full Stack Web Development

Course Code: 23CM4121

Experiment No.: 1

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1. TITLE

Develop a College Information Portal Using HTML5

2. AIM

To develop a static web page — the **College Information Portal** — using **HTML5**, demonstrating text formatting elements, ordered and unordered lists, hyperlinks, tabular data, image and multimedia embedding, and HTML5 form input types.

3. OBJECTIVE

The objectives of this experiment are:

- To understand the basic structure of an HTML5 document and its semantic elements.
 - To create a structured web page using `<header>`, `<nav>`, `<section>`, `<article>`, and `<footer>`.
 - To apply inline text formatting using `<b>`, `<i>`, `<u>`, `<sub>`, and `<sup>`.
 - To create ordered lists `<ol>`, unordered lists `<ul>`, and hyperlinks `<a>`.
 - To display tabular data using `<table>`, `<tr>`, `<th>`, and `<td>`.
 - To embed multimedia using `<img>`, `<audio>`, and `<video>` tags.
 - To build a registration form using HTML5 input types: text, email, password, date, number, color, range, radio, checkbox, select, and textarea.
-

4. THEORY

HTML5 is the fifth and current major version of HyperText Markup Language, the standard markup language used to structure and present content on the World Wide Web. It introduced new semantic elements, native multimedia support, and a richer set of form input types, reducing the dependency on external plug-ins.

Semantic Elements such as `<header>`, `<nav>`, `<section>`, `<article>`, and `<footer>` give meaning to the structure of the page, improving readability, maintainability, accessibility, and search-engine indexing.

Text Formatting Tags — `<b>` for bold, `<i>` for italics, `<u>` for underline, `<sub>` for subscript (H_2O) and `<sup>` for superscript ($E = mc^2$) — control the appearance of inline text.

Lists and Hyperlinks — `<ol>` produces a numbered list, `<ul>` produces a bulleted list, and `<a href>` creates a hyperlink. The `target="_blank"` attribute opens the linked page in a new browser tab.

Tables organise data in rows and columns. `<tr>` defines a row, `<th>` a heading cell, and `<td>` a data cell. The `border` and `cellpadding` attributes control the ruling and the spacing inside each cell.

Multimedia Tags — `<img>`, `<audio>` and `<video>` — allow images, audio and video content to be embedded directly in the browser. The `controls` attribute displays the native play, pause and volume controls, and the fallback text inside the element is shown when the browser cannot play the media.

In this page, HTML5 forms are used to collect student registration data.

- `type="text"` accepts a single line of plain text.
- `type="email"` validates the email format in the browser.
- `type="password"` masks the characters as they are typed.
- `type="date"` provides a built-in date picker.
- `type="number"` restricts input to numeric values within a range.
- `type="color"` opens a colour picker widget.
- `type="range"` renders a slider control between a minimum and maximum value.
- `type="radio"` allows selection of exactly one option from a group.
- `type="checkbox"` allows multiple selections from a group.
- `<select>` provides a dropdown list and `<textarea>` accepts multi-line text.

The basic structure of the page is:

Header → Navigation → Main Content (About, Students, Registration, Media) → Footer

5. ALGORITHM / PROCEDURE

- Create a new file named `College_Information_Portal.html`.
- Add the `<!DOCTYPE html>` declaration.
- Add the `<html lang="en">` root element.
- Inside `<head>`, add the `charset` meta tag, the `viewport` meta tag, and the page title.
- Create a `<header>` containing the portal heading `<h1>` and a welcome line.
- Create a `<nav>` bar with internal hyperlinks to the About, Students, Registration and Media sections.
- Add a `<hr>` separator after the navigation bar.
- Create the "About Our College" `<section id="about">` containing an `<article>`.
- Use `<b>`, `<i>`, `<u>`, `<sub>` and `<sup>` for inline formatting within the article.
- Add an ordered list `<ol>` for Courses Offered and an unordered list `<ul>` for College Facilities.
- Add external hyperlinks with `target="_blank"` under Useful Links.
- Create the "Student Details" `<section id="students">` with a bordered `<table>` having four heading cells and three student rows.
- Add the College Logo and Campus Image using `<img>` tags with `src`, `alt` and `width` attributes.
- Create the "Student Registration Form" `<section id="registration">` with a `<form>`.

- Inside the form add text, email, password, radio, checkbox, select, textarea, submit and reset controls.
 - Create the "Media" <section id="media"> with the HTML5 input types email, date, number, color and range.
 - Embed audio using <audio controls> and video using <video controls> with their <source> elements.
 - Add a <footer> with a copyright notice.
 - Save the file and open it in a browser to verify the output.
-

6. PROGRAM

College_Information_Portal.html

```
<!DOCTYPE html>
<html lang="en">

<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>College Information Portal</title>
</head>
<body>

  <!-- ===== -->
  <!-- HEADER SECTION -->
  <!-- ===== -->
  <header>
    <h1>College Information Portal</h1>
    <p>Welcome to the official student information portal.</p>
  </header>

  <!-- ===== -->
  <!-- NAVIGATION BAR -->
  <!-- ===== -->
  <nav>
    <a href="#about">About</a> |
    <a href="#students">Students</a> |
    <a href="#registration">Registration</a> |
    <a href="#media">Media</a>
  </nav>

  <hr>

  <!-- ===== -->
  <!-- PART A : BASIC HTML -->
  <!-- ===== -->
  <section id="about">

    <article>
      <h2>About Our College</h2>

      <p>
        <b>ANITS</b> (Anil Neerukonda Institute of Technology & Sciences)
        is one of the leading engineering colleges in Andhra Pradesh.
      </p>

      <p>
        <i>The college provides quality education in Engineering,
        Technology and Management.</i>
      </p>

      <p>
        <u>Our vision is to nurture future innovators.</u>
      </p>

      <p>
        Water Formula:  $H<sub>2</sub></math>
      </p>

      <p>
        Einstein's Equation:  $E = mc<sup>2</sup></math>
      </p>
    </article>

  </section>$$ 
```

```

<!-- Ordered List -->
<h3>Courses Offered</h3>

<ol>
  <li>Computer Science Engineering</li>
  <li>Artificial Intelligence & Machine Learning</li>
  <li>Electronics & Communication Engineering</li>
  <li>Mechanical Engineering</li>
</ol>

<!-- Unordered List -->
<h3>College Facilities</h3>

<ul>
  <li>Central Library</li>
  <li>Computer Labs</li>
  <li>Sports Complex</li>
  <li>Hostel Facility</li>
</ul>

<!-- Hyperlinks -->
<h3>Useful Links</h3>

<a href="https://www.anits.edu.in" target="_blank">
  Visit College Website
</a>

<br><br>

<a href="https://www.google.com" target="_blank">
  Visit Google
</a>

</section>

<hr>

<!-- ===== -->
<!-- PART B : TABLES & IMAGES -->
<!-- ===== -->
<section id="students">

  <h2>Student Details</h2>

  <table border="1" cellpadding="8">

    <tr>
      <th>Roll No</th>
      <th>Name</th>
      <th>Branch</th>
      <th>Email</th>
    </tr>

    <tr>
      <td>101</td>
      <td>Sai Shishir</td>
      <td>CSE (AI & ML)</td>
      <td>shishir@gmail.com</td>
    </tr>

    <tr>
      <td>102</td>
      <td>Rahul</td>
      <td>CSE</td>
      <td>rahul@gmail.com</td>
    </tr>

    <tr>
      <td>103</td>
      <td>Priya</td>
      <td>ECE</td>
      <td>priya@gmail.com</td>
    </tr>

  </table>

  <br>

  <h2>College Logo</h2>

  <br><br>

  <h2>Campus Image</h2>

  
</section>
<hr>
<!-- ===== -->
<!-- PART C : HTML FORMS -->
<!-- ===== -->
<section id="registration">
    <h2>Student Registration Form</h2>
    <form>
        <label>Name:</label><br>
        <input type="text" name="name"><br><br>
        <label>Email:</label><br>
        <input type="email" name="email"><br><br>
        <label>Password:</label><br>
        <input type="password" name="password"><br><br>
        <label>Gender:</label><br>
        <input type="radio" name="gender"> Male
        <input type="radio" name="gender"> Female
        <input type="radio" name="gender"> Other
        <br><br>
        <label>Hobbies:</label><br>
        <input type="checkbox"> Reading
        <input type="checkbox"> Sports
        <input type="checkbox"> Music
        <br><br>
        <label>Branch:</label><br>
        <select>
            <option>CSE</option>
            <option>CSE (AI & ML)</option>
            <option>ECE</option>
            <option>EEE</option>
            <option>Mechanical</option>
        </select>
        <br><br>
        <label>Comments:</label><br>
        <textarea rows="5" cols="40"></textarea>
        <br><br>
        <input type="submit" value="Submit">
        <input type="reset" value="Reset">
    </form>
</section>
<hr>
<!-- ===== -->
<!-- PART D : HTML5 FEATURES -->
<!-- ===== -->
<section id="media">
    <h2>HTML5 Input Types</h2>
    <form>
        <label>Email:</label><br>
        <input type="email"><br><br>
        <label>Date of Birth:</label><br>
        <input type="date"><br><br>
        <label>Age:</label><br>
        <input type="number"><br><br>
        <label>Favorite Color:</label><br>
        <input type="color"><br><br>
        <label>Satisfaction Level:</label><br>
        <input type="range" min="0" max="100">
    </form>
<br>

```

```

        <h2>Audio</h2>
        <audio controls>
            <source src="college_song.mp3" type="audio/mpeg">
            Your browser does not support audio.
        </audio>
        <br><br>
        <h2>Video</h2>
        <video width="400" controls>
            <source src="college_video.mp4" type="video/mp4">
            Your browser does not support video.
        </video>
    </section>
    <hr>
    <!-- ===== -->
    <!-- FOOTER SECTION -->
    <!-- ===== -->
    <footer>
        <p>
            &copy; 2026 College Information Portal |
            All Rights Reserved.
        </p>
    </footer>
</body>
</html>

```

7. CODE EXPLANATION

Code	Explanation
<html>	Defines the HTML document.
<head>	Contains webpage information, meta tags and the title.
<body>	Contains the visible webpage content.
<header>	Defines the header of the portal.
<nav>	Defines the navigation bar of the portal.
<a>	Creates internal and external hyperlinks.
<section>	Groups related content together.
<article>	Contains the college overview information.
<h1> - <h3>	Define headings of different levels.
<p>	Defines a paragraph of text.
 / <i> / <u>	Apply bold, italic and underline formatting.
<sub> / <sup>	Display subscript and superscript text.
/	Create ordered and unordered lists.
	Defines an individual list item.
<table>	Displays student details in tabular form.
<tr>	Defines a row of the table.
<th>	Defines table heading cells.

Code	Explanation
<td>	Defines table data cells.
	Displays the college logo and campus image.
<audio>	Adds audio to the webpage.
<video>	Adds video to the webpage.
<form>	Creates the student registration form.
<label>	Labels each form control.
<input>	Accepts different types of student information.
<select>	Creates a dropdown for branch selection.
<textarea>	Allows users to enter comments.
<hr>	Draws a horizontal separator line.
<footer>	Defines the footer of the portal.

8. TEST CASES AND EXECUTION DATA

Test Case	Input / Action	Expected Result	Actual Result	Status
TC-01	Open the HTML file in a web browser	Static webpage should be displayed correctly	Static webpage displayed correctly	Pass
TC-02	Click the navigation links (About, Students, Registration, Media)	Page should scroll to the corresponding section	Page scrolled to the section	Pass
TC-03	Click the "Visit College Website" link	Official college website should open in a new tab	College website opened in a new tab	Pass
TC-04	Verify the Student Details table	Three student records should be displayed with borders	Table displayed correctly	Pass
TC-05	Enter valid name, email, password and comments in the form	Form should accept the entered information	Information accepted	Pass
TC-06	Enter an invalid value in the Email field and submit	Browser should display email format validation	Validation message displayed	Pass
TC-07	Click the Reset button	All form fields should be cleared	All form fields are reset	Pass
TC-08	Use the date, number, color and range input controls	Date picker, spinner, colour picker and slider should appear	All HTML5 controls rendered correctly	Pass
TC-09	Play the embedded audio and video	Media controls should be displayed by the browser	Media controls displayed	Pass

9. OUTPUT

Output 1: Header, Navigation, About Section, Lists, Links and Student Details Table

8/4/26, 12:10 PM

College Information Portal

College Information Portal

Welcome to the official student information portal.

[About](#) | [Students](#) | [Registration](#) | [Media](#)

About Our College

ANITS (Anil Neerukonda Institute of Technology & Sciences) is one of the leading engineering colleges in Andhra Pradesh.

The college provides quality education in Engineering, Technology and Management.

Our vision is to nurture future innovators.

Water Formula: H_2O

Einstein's Equation: $E = mc^2$

Courses Offered

1. Computer Science Engineering
2. Artificial Intelligence & Machine Learning
3. Electronics & Communication Engineering
4. Mechanical Engineering

College Facilities

- Central Library
- Computer Labs
- Sports Complex
- Hostel Facility

Useful Links

[Visit College Website](#)

[Visit Google](#)

Student Details

Roll No	Name	Branch	Email
101	Sai Shishir	CSE (AI & ML)	shishir@gmail.com
102	Rahul	CSE	rahul@gmail.com

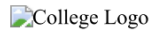
Output 2: Images, Student Registration Form and HTML5 Input Types

8/4/26, 12:10 PM

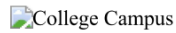
College Information Portal

103	Priya	ECE	priya@gmail.com
-----	-------	-----	-----------------

College Logo



Campus Image



Student Registration Form

Name:

Email:

Password:

Gender:

☐ Male ☐ Female ☐ Other

Hobbies:

☐ Reading ☐ Sports ☐ Music

Branch:

Comments:

HTML5 Input Types

Email:

Date of Birth:

Output 3: HTML5 Input Types, Audio, Video and Footer

8/4/26, 12:10 PM

College Information Portal

Age:

Favorite Color:

Satisfaction Level:

Audio

0:00 / 0:00

Video

0:00

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10. RESULT

Thus, a structured static webpage — the College Information Portal — was successfully designed and developed using HTML5 semantic elements and HTML components. The webpage was verified for its structure, navigation links, text formatting, ordered and unordered lists, tabular data, embedded images, audio and video, form controls, HTML5 input types and footer, and all test cases were executed successfully.